

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Analyzed experimental data and presented findings to technical teams and scientific meetings, leading to a **15%** improvement in assay validation success rates.
- Developed and supported system-level requirements aligned with product goals, ensuring compliance with **good laboratory practice (GLP)** standards for 10+ development projects.
- Investigated, troubleshot, and resolved technical issues in reagent systems and instruments, reducing critical bug resolution time by **20%** through structured root-cause analysis.
- Designed and executed complex experiments and test strategies for reagent systems and instruments, achieving **98%** data integrity and reproducibility.
- Authored and maintained clear, accurate protocols, technical reports, and study summaries, improving documentation consistency by **30%** across the R&D; department.
- Collaborated with cross-functional teams including R&D;, Manufacturing, and Quality Assurance to meet program achievements and commitments, contributing to the successful launch of 3 new product lines.

SKILLS

Scientific Expertise – Experimental Planning, Data Analysis, Troubleshooting, Scientific Judgment, Critical Thinking, GLP Compliance

Laboratory Techniques – Reagent Systems, Assays, Workflows, Instruments, Protocol Development, Study Summaries

Data & Analytics – Statistical Analysis, Data Interpretation, R, Python (pandas, NumPy, scikit-learn), Power BI, DAX

Software & Tools – Jira, Confluence, Splunk, Dynatrace, Jenkins, Git, Linux, AWS (EC2/S3/Lambda/CloudWatch), Docker

Programming & Systems – Java (Spring Boot), Python, SQL (PostgreSQL, MySQL, Oracle), REST APIs, ERP Systems Integration

Domain Knowledge – Life Sciences, Medical Technology, Regulated Environments, Customer Workflows, Regulatory Requirements

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Automated Assay Performance Monitoring System (HCL Technologies)

- Developed an automated system using **Python** and **Flask** to monitor the performance of critical assays in a regulated laboratory environment, processing over **1,000** data points daily.
- Integrated with laboratory instruments via **REST APIs** to collect real-time performance metrics, achieving a **99.5%** data capture success rate and enabling proactive issue identification.
- Built a **Power BI** dashboard to visualize assay trends, identify deviations, and provide actionable insights to R&D; scientists, reducing manual data review time by **40%**.

Real-Time Transactional Data Analysis Pipeline (HCL Technologies)

- Engineered a data pipeline using **Java Spring Boot** and **PostgreSQL** to ingest and process high-volume financial transaction messages, supporting a payments operations dashboard.
- Implemented robust monitoring and alerting mechanisms with **Splunk** for transaction flow anomalies and reconciliation workflows, maintaining **99.9%** operational uptime.
- Supported cross-functional teams in troubleshooting and triaging operational issues, ensuring adherence to strict SLAs in a GDPR-aware production environment.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)